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3rd Year Physics Major
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Research Proposal:
Levitation and Dynamics of Water Particles by Thermophoretic force
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Abstract This project is a continuation of successful work conducted by Nicholas Kowalski and Frankie Fung. The project centers around trapping water particles in a low-pressure environment using the thermophoretic force [1]. The current experimental setup is capable of levitating ice particles as described by Kowalski et al. [2], but stable levitation of water particles remains unachieved. The proposed project will aim to achieve stable levitation of water droplets and study their dynamics, as well as providing undergraduate Misha Usatyuk, an opportunity to expand his research experience and develop as a Physics major. The research the student will be involved in over the summer of 2016, focusing on the dynamics of levitating water particles, will utilize and expand his knowledge of physics.

The foundation of this experiment lies in the thermophoretic force. When a particle is placed in a low pressure gaseous environment with a temperature gradient, thermophoretic forces are produced. The dynamics induced by the thermal gradient are called thermophoresis. The particles initially levitate a few millimeters above a hot plate and then accelerate toward a cold plate above the particles by the thermophoretic force [1]. In the appropriate pressure and temperature regime this allows for stable levitation of ice particles for durations of an hour or longer. We are currently able to levitate and study the dynamics of ice particles, but levitation of water particles remains elusive. According to preliminary theory such levitation is possible with some modification to the experimental apparatus.

Currently, a setup exists that creates a temperature gradient at around torr pressures. The setup uses a TE cooler and a liquid nitrogen bath to create temperatures of 250 K and 77K at the bottom and the top of a vacuum chamber [2]. A similar setup exists with the Peltier cooler replaced by a heater, and we are capable of heating the copper plate to any needed temperature. This temperature gradient, along with the low pressure maintained by a pump, is used to create appropriate thermophoretic forces for the levitation of water particles. However, this levitation is not stable.

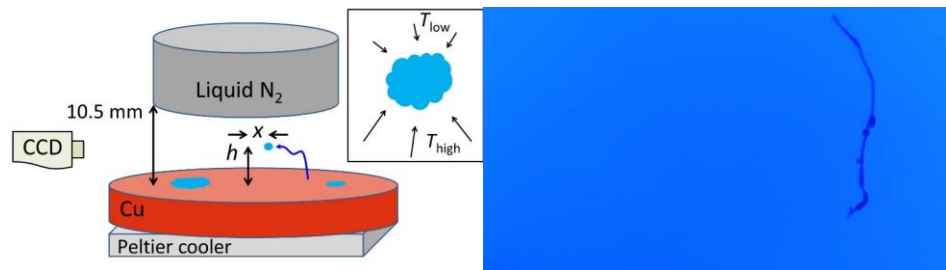


Figure 1&2. Experimental setup of the ice particle levitation system. The system, built by N. Kowalski and coworkers during Summer 2014, consists of a hot plate controlled by a Peltier cooler at 270 K and a cold plate at 77K cooled by liquid Nitrogen. Ice particles are levitated by thermophoretic force and are monitored by a CCD camera. An identical setup with the Peltier cooler replaced by a heater is functional and can be adapted to levitate water particles. Figure 2 demonstrates the power and flexibility of our experimental setup with the levitation of an object approximately 1 mm in size.

Due to the thermophoretic force being strongest at lower pressure and the behavior of water at low pressure, we need to achieve levitation of the water particles at a narrow range of temperatures. This can be achieved by heating the copper plate as described by figure 3.

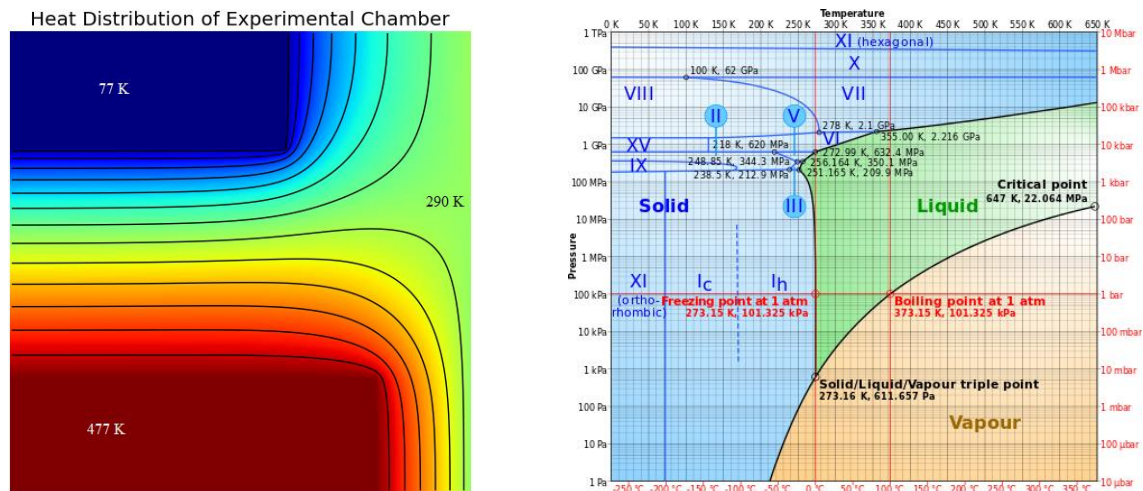


Figure 3&4. Possible setup to levitate water particles (numerical simulation). The center between the two plates is around 277 Kelvin, which corresponds to liquid water at torr pressures. The contours represent surfaces of constant temperature. The phase diagram is present to indicate the regime in which liquid water forms. The experimental apparatus operates around 10-25 torr.

The current issue with the above is an instability of the thermophoretic force pointing towards the sides of the chamber. This occurs when the bottom plate is heated above the ambient temperature of the room, but we believe this can be fixed by adjusting the apparatus.

Science goals for the summer are twofold: we will aim to establish stable levitation of liquid water particles, and we will observe the levitation dynamics of such particles similar to previous work done on ice particles [2]. There also remains the possibility of pushing the levitating water particles with a laser, and observing the resulting behavior.

Our expectations for this project are to successfully levitate water particles and explore their dynamics. This will allow us to achieve a new method of trapping water droplets, and could open up a new technique to be used in various avenues of research. These techniques will have interesting applications to research in chemistry, molecular, atmospheric, and even astro-physics.

References:

- [1] T. Kelling, G. Wurm, and C. Dürmann, “Ice Particles Trapped by Temperature Gradients at mbar Pressure” *Review of Scientific Instruments* **82**, (2011)
- [2] N. Kowalski, B. Xie, C. V. Parker, and C. Chin, “Stable levitation and dynamics of ice particles at low pressures” arXiv:1504.01035v2 (2015), submitted to PRX
- [3] https://en.wikipedia.org/wiki/Phase_diagram#Crystal_phase_diagrams